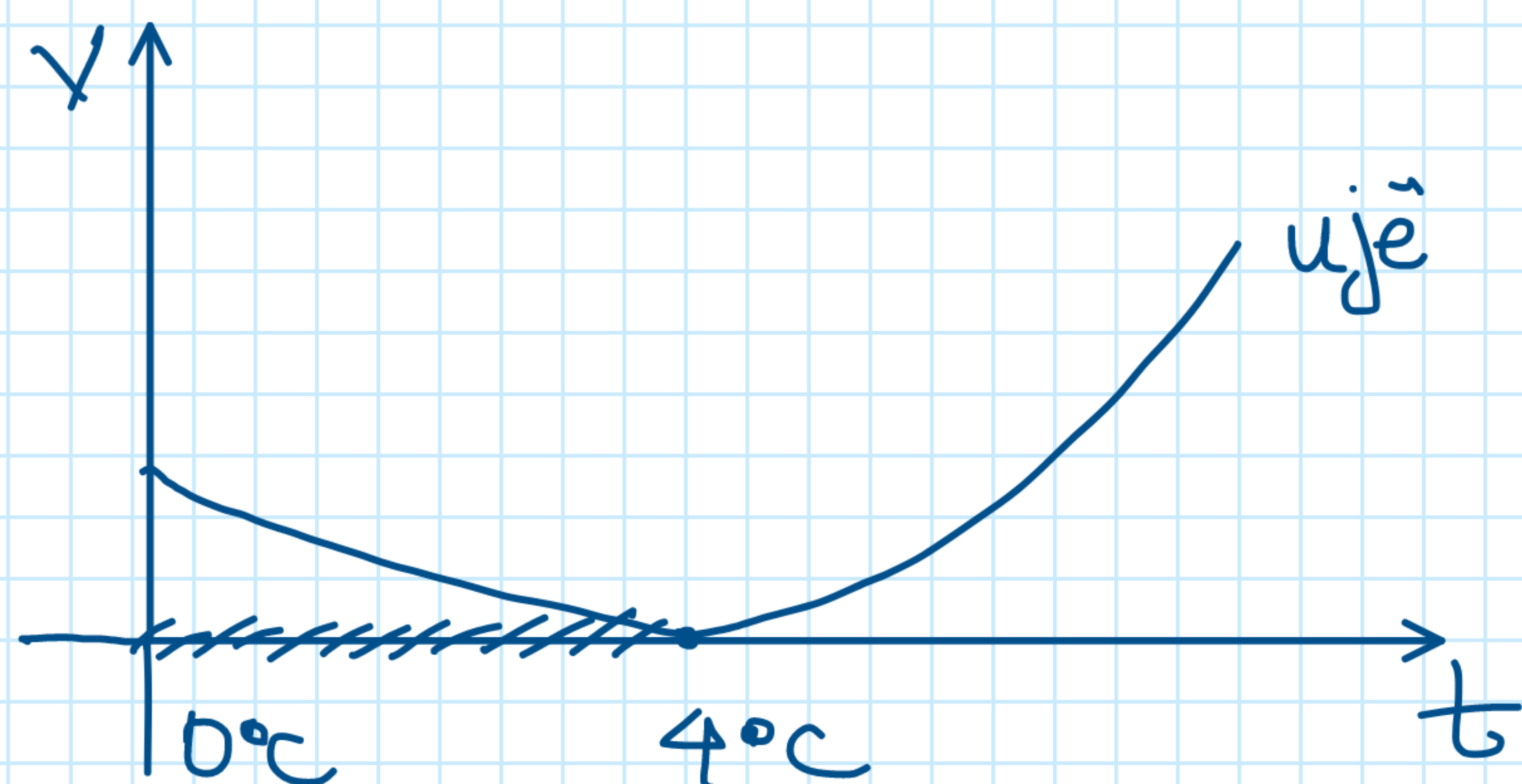


Ligjerata 10dh

Anomalia e zgjerimit të ujit



Tensioni mekanik termik i trupave të ngurtë

$$l_1 = l_0 (1 + \alpha t_1)$$

$$l_2 = l_0 (1 + \alpha t_2)$$

$$\Delta l = l_2 - l_1 = \cancel{l_0} + l_0 \alpha t_2 - \cancel{l_0} - l_0 \alpha t_1$$

$$\Delta l = l_0 \alpha (t_2 - t_1)$$

$$\Delta t = t_2 - t_1$$

$$\Delta l = \alpha \cdot l_0 \Delta t \quad *$$

$$\Delta l = k l_0 \sigma \quad **$$

$$\sigma = \frac{F}{S}$$

$$k = \frac{1}{E}$$

$$k l_0 \sigma = \alpha l_0 \Delta t$$

$$\sigma = \frac{\alpha \cdot \Delta t}{\frac{1}{E}} = \alpha E \cdot \Delta t$$

$$\sigma = \alpha \cdot E \cdot \Delta t$$